

TT1 · Section A · S460 transfer truss

HK Steel Code 2011 (2023 edition), April 2026 amendments. OK within stated scope.

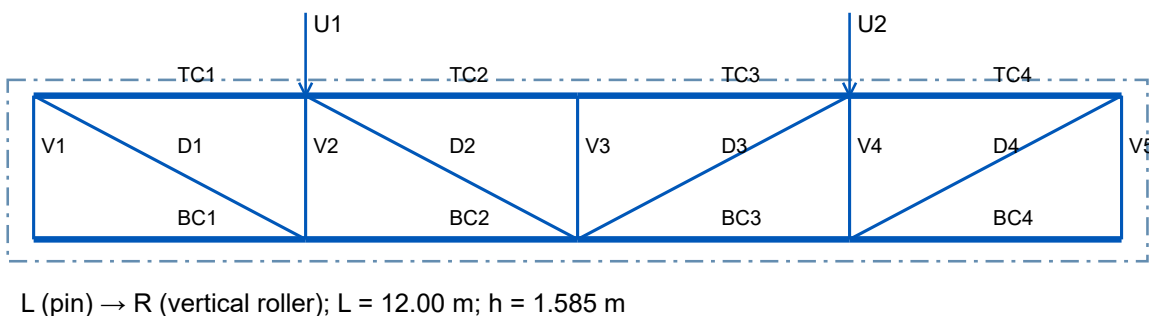
Bottom 3/F $z=10.24$ m; top 3/F $z=11.83$ m. Upper column segment: 4/F.

Same-floor Structural zone: outside depth 2000.00 mm, full-span available 2000.00 mm. Fixed floor $z=12.00$ m; chord axis spacing 1585.00 mm.

Top / bottom axis setbacks = 174.00 / 241.00 mm; includes chord half-depths and any larger diagonal vertical projection.

$E = 205,000$ N/mm²; S460 $p_y = 460 / 440 / 430 / 410 / 400$ MPa for $t \leq 16 / 40 / 63 / 80 / 100$ mm (Table 3.2).

Planar concentric pin-jointed axial model; BS hot-rolled UB/UC; steel alone. Joint concrete adds weight only. No weld / rebar anchorage / connection / fire / whole-truss torsion design. Member SW lumped at nodes; local bending requires separate review.



Characteristic reactions: L $G=1108.41$, $Q=450.00$ kN; R $G=808.41$, $Q=750.00$ kN.

Steel SW = 87.41 kN; additional joint concrete = 29.40 kN. Reactions enter same-floor supporting columns, then accumulate downwards.

Deflection envelopes $G+Q / Q /$ differential = 17.47 / 6.77 / 2.31 mm; project limits = 20 / 12 / 8 mm.

In-plane stiffness remains positive with 10× compression envelope: YES. Out-of-plane lateral restraint assumption: confirmed in model.

TT1 · Section A · S460 transfer truss · Loads & method

Separate characteristic G and Q; no reuse of already-factored column design load.

ULS envelope: each G uses adverse 1.4 / beneficial 1.0; each independent Q uses adverse 1.6 / beneficial 0 (Table 4.2). Each member has its own governing arrangement.

Solve $K u = P$ with EA/L bar stiffness. $N = EA/L \times$ axial displacement. Both chord levels and web axial deformation are included.

$P_t = A_g p_y$ (8.6.1, no holes assumed). $P_c = A_g \min(p_{cx}, p_{cy})$ (8.7, Appendix 8.4). $\epsilon = \sqrt{(275/p_y)}$; conservative flange $B/(2t_f) \leq 13\epsilon$; web $d/t_w \leq 40\epsilon$. Class 4 compression excluded.

$p_c = p_y p_E / [\phi + \sqrt{(\phi^2 - p_y p_E)}]$; $p_E = \pi^2 E / \lambda^2$; $\phi = [p_y + (1 + \eta)p_E]/2$; $\eta = \max[0, \alpha(\lambda - \lambda_0)/1000]$; $\lambda_0 = 0.2\pi\sqrt{(E/p_y)}$.

Rolled UB curves x/y: a/b ($t_f \leq 40$), b/c (> 40). UC: b/c, c/d. $\alpha_a = 2$, $\alpha_b = 3.5$, $\alpha_c = 5.5$, $\alpha_d = 8$. $\lambda_c \leq 200$; $\lambda_t \leq 300$. No built-up welded-section p_y reduction applied.

Source / position	G / Q (kN)	Additional joint volume (m³)
U1 @ 3.00 m	1200.00 / 300.00	0.40
U2 @ 9.00 m	600.00 / 900.00	0.40
Support joints	24.5 kN/m³ × volume	0.20 / 0.20

Equilibrium residuals: $\Sigma V = 5.68e-12$ kN; $\Sigma M = 4.37e-11$ kNm.

TT1 · A · Force solution · Nodes

Planar pin-jointed axial members. Member-end $M = V = 0$ in this idealisation; local bending from loads between joints is outside scope.

Origin: left bottom chord node. x right / y up; member N positive in tension. Nodal G / Q below are positive DOWN, including steel and additional joint concrete.

Left bottom: $u_x = u_y = 0$. Right bottom: $u_y = 0$, u_x free. Solve 17 free DOF; load cases are solved independently before combination.

Node / x, y (m)	G / Q downward (kN)	ULS P_y upward (kN)
B0 0.000 / 0.000	8.495 / 0.000	-11.892
B1 3.000 / 0.000	9.958 / 0.000	-13.941
B2 6.000 / 0.000	13.718 / 0.000	-19.205
B3 9.000 / 0.000	9.958 / 0.000	-13.941
B4 12.000 / 0.000	8.495 / 0.000	-11.892
T0 0.000 / 1.585	8.075 / 0.000	-11.305
T1 3.000 / 1.585	1221.200 / 300.000	-2189.680
T2 6.000 / 1.585	7.640 / 0.000	-10.696
T3 9.000 / 1.585	621.200 / 900.000	-2309.680
T4 12.000 / 1.585	8.075 / 0.000	-11.305

ULS column uses $1.4G + 1.6Q$ with all variable loads present. Individual member envelopes may use different load patterns.

TT1 · A · Force solution · Reactions & stiffness

Worked load case: 1.4G + 1.6Q, all loads present.

$$\Sigma F_x = 0: R_{x,L} = -\Sigma P_x = 0.000 \text{ kN}.$$

$$\Sigma M_{,L} = 0: R_{y,R} = -\Sigma(x P_y - y P_x) / L = -(-27981.232) / 12.000 = 2331.769 \text{ kN}.$$

$$\Sigma F_y = 0: R_{y,L} = -\Sigma P_y - R_{y,R} = -(-4603.539) - 2331.769 = 2271.769 \text{ kN}.$$

$$D1: L = 3.392967 \text{ m}; c = 0.884182; s = -0.467143; A = 29000 \text{ mm}^2.$$

$$EA/L = 205000 \times 29000 / (1000 \times 3.392967) = 1752153.798 \text{ kN/m}.$$

$$v = [-c, -s, c, s]; k_e = (EA/L) v^T v; \text{ assemble } K_{ff} u_f = P_f \text{ (17 free DOF)}.$$

$$\Delta L = c(u_{x,j} - u_{x,i}) + s(u_{y,j} - u_{y,i}) = 2.747166e-3 \text{ m}.$$

$$N = (EA/L) \Delta L = 4813.457 \text{ kN}; + \text{ tension} / - \text{ compression}.$$

$K_{ff} = L L^T$; solve $L y = P_f$, then $L^T u_f = y$. Recover member N from axial extension; recover reactions at restrained DOF.

Free DOF order: B1.ux, B1.uy, B2.ux, B2.uy, B3.ux, B3.uy, B4.ux, T0.ux, T0.uy, T1.ux, T1.uy, T2.ux, T2.uy, T3.ux, T3.uy, T4.ux, T4.uy

Independent method-of-joints verification: $\max |N_{,joints} - N_{,stiffness}|$ across G, Q, combinations and individual actions = $1.182e-11 \text{ kN}$.

TT1 · A · Force solution · Axial forces 1

All axial forces in kN: + tension / – compression. G and Q include all respective characteristic loads.

Full-load ULS = 1.4 NG + 1.6 NQ. Envelope extrema independently select G factors 1.0/1.4 and Q factors 0/1.6 by sign of each action.

Member / length (m)	NG / NQ (kN)	Full ULS; envelope min / max
BC1 L=3.000	0.000 / 0.000	0.000 0.000 / 0.000
TC1 L=3.000	-2066.568 / -851.735	-4255.971 -4255.971 / -2066.568
D1 L=3.393	2337.266 / 963.303	4813.457 2337.266 / 4813.457
BC2 L=3.000	2066.568 / 851.735	4255.971 2066.568 / 4255.971
TC2 L=3.000	-1802.869 / -1135.647	-4341.051 -4341.051 / -1802.869
D2 L=3.393	-298.241 / 321.101	96.224 -812.002 / 609.986
BC3 L=3.000	1498.745 / 1419.558	4369.536 1498.745 / 4369.536
TC3 L=3.000	-1802.869 / -1135.647	-4341.051 -4341.051 / -1802.869

Do not combine separate member envelope extrema into one equilibrium load case.

TT1 · A · Force solution · Axial forces 2

All axial forces in kN: + tension / – compression. G and Q include all respective characteristic loads.

Full-load ULS = 1.4 NG + 1.6 NQ. Envelope extrema independently select G factors 1.0/1.4 and Q factors 0/1.6 by sign of each action.

Member / length (m)	NG / NQ (kN)	Full ULS; envelope min / max
D3 L=3.393	343.961 / -321.101	-32.216 -555.122 / 866.867
BC4 L=3.000	0.000 / 0.000	0.000 0.000 / 0.000
TC4 L=3.000	-1498.745 / -1419.558	-4369.536 -4369.536 / -1498.745
D4 L=3.393	1695.064 / 1605.505	4941.897 1695.064 / 4941.897
V1 L=1.585	-1099.912 / -450.000	-2259.877 -2259.877 / -1099.912
V2 L=1.585	-1081.879 / -450.000	-2234.630 -2234.630 / -1081.879
V3 L=1.585	-7.640 / 0.000	-10.696 -10.696 / -7.640
V4 L=1.585	-781.879 / -750.000	-2294.630 -2294.630 / -781.879

Do not combine separate member envelope extrema into one equilibrium load case.

TT1 · A · Force solution · Axial forces 3

All axial forces in kN: + tension / – compression. G and Q include all respective characteristic loads.

Full-load ULS = 1.4 NG + 1.6 NQ. Envelope extrema independently select G factors 1.0/1.4 and Q factors 0/1.6 by sign of each action.

Member / length (m)	NG / NQ (kN)	Full ULS; envelope min / max
V5 L=1.585	-799.912 / -750.000	-2319.877 -2319.877 / -799.912

Do not combine separate member envelope extrema into one equilibrium load case.

TT1 · A · Force solution · Joint equilibrium 1

Independent node-by-node solution for the full-load ULS case. Known member forces and external loads (including support reactions) are moved to the right-hand side. T = tension, C = compression.

Step / node; external Fx, Fy	Unknowns from $\Sigma F_x = \Sigma F_y = 0$	Solved forces (kN)
1 · B0 Fx 0.000 Fy 2259.877	$\Sigma F_x: 1.000000 \text{ N(BC1)} + 0.000000$ $\text{N(V1)} = 0.000$ $\Sigma F_y: 0.000000 \text{ N(BC1)} + 1.000000$ $\text{N(V1)} = -2259.877$	$\text{N(BC1)} = 0.000 \text{ kN (zero)}$ $\text{N(V1)} = -2259.877 \text{ kN (C)}$
2 · B4 Fx 0.000 Fy 2319.877	$\Sigma F_x: -1.000000 \text{ N(BC4)} + 0.000000$ $\text{N(V5)} = 0.000$ $\Sigma F_y: 0.000000 \text{ N(BC4)} + 1.000000$ $\text{N(V5)} = -2319.877$	$\text{N(BC4)} = 0.000 \text{ kN (zero)}$ $\text{N(V5)} = -2319.877 \text{ kN (C)}$
3 · T0 Fx 0.000 Fy -11.305	$\Sigma F_x: 1.000000 \text{ N(TC1)} + 0.884182$ $\text{N(D1)} = 0.000$ $\Sigma F_y: 0.000000 \text{ N(TC1)} + -0.467143$ $\text{N(D1)} = -2248.572$	$\text{N(TC1)} = -4255.971 \text{ kN (C)}$ $\text{N(D1)} = 4813.457 \text{ kN (T)}$
4 · T4 Fx 0.000 Fy -11.305	$\Sigma F_x: -1.000000 \text{ N(TC4)} + -0.884182$ $\text{N(D4)} = 0.000$ $\Sigma F_y: 0.000000 \text{ N(TC4)} + -0.467143$ $\text{N(D4)} = -2308.572$	$\text{N(TC4)} = -4369.536 \text{ kN (C)}$ $\text{N(D4)} = 4941.897 \text{ kN (T)}$
5 · B1 Fx 0.000 Fy -13.941	$\Sigma F_x: 1.000000 \text{ N(BC2)} + 0.000000$ $\text{N(V2)} = 4255.971$ $\Sigma F_y: 0.000000 \text{ N(BC2)} + 1.000000$ $\text{N(V2)} = -2234.630$	$\text{N(BC2)} = 4255.971 \text{ kN (T)}$ $\text{N(V2)} = -2234.630 \text{ kN (C)}$

Maximum residual over ALL node equilibrium equations: 1.819e-12 kN.

TT1 · A · Force solution · Joint equilibrium 2

Independent node-by-node solution for the full-load ULS case. Known member forces and external loads (including support reactions) are moved to the right-hand side. T = tension, C = compression.

Step / node; external Fx, Fy	Unknowns from $\Sigma F_x = \Sigma F_y = 0$	Solved forces (kN)
6 · B3 Fx 0.000 Fy -13.941	$\Sigma F_x: -1.000000 \text{ N(BC3)} + 0.000000$ $\text{N(V4)} = -4369.536$ $\Sigma F_y: 0.000000 \text{ N(BC3)} + 1.000000$ $\text{N(V4)} = -2294.630$	$\text{N(BC3)} = 4369.536 \text{ kN (T)}$ $\text{N(V4)} = -2294.630 \text{ kN (C)}$
7 · T1 Fx 0.000 Fy -2189.680	$\Sigma F_x: 1.000000 \text{ N(TC2)} + 0.884182$ $\text{N(D2)} = -4255.971$ $\Sigma F_y: 0.000000 \text{ N(TC2)} + -0.467143$ $\text{N(D2)} = -44.951$	$\text{N(TC2)} = -4341.051 \text{ kN (C)}$ $\text{N(D2)} = 96.224 \text{ kN (T)}$
8 · T2 Fx 0.000 Fy -10.696	$\Sigma F_x: 1.000000 \text{ N(TC3)} + 0.000000$ $\text{N(V3)} = -4341.051$ $\Sigma F_y: 0.000000 \text{ N(TC3)} + -1.000000$ $\text{N(V3)} = 10.696$	$\text{N(TC3)} = -4341.051 \text{ kN (C)}$ $\text{N(V3)} = -10.696 \text{ kN (C)}$
9 · T3 Fx 0.000 Fy -2309.680	$\Sigma F_x: -0.884182 \text{ N(D3)} = 28.485$ $\Sigma F_y: -0.467143 \text{ N(D3)} = 15.049$	$\text{N(D3)} = -32.216 \text{ kN (C)}$

Maximum residual over ALL node equilibrium equations: 1.819e-12 kN.

TT1 · A · Force solution · Governing envelope

Governing member BC3: $N_{min} = \sum \gamma_{min,i} N_i = 1498.745 \text{ kN}$; $N_{max} = \sum \gamma_{max,i} N_i = 4369.536 \text{ kN}$.

Compression demand = $\max(0, -N_{min})$; tension demand = $\max(0, N_{max})$. Each other member is enveloped in the same way using its own action signs.

Independent action	Characteristic N (kN)	$\gamma_{min} / \gamma_{max}$
Steel self-weight	60.638	1.0 / 1.4
Joint concrete (additional)	18.549	1.0 / 1.4
U1 G	567.823	1.0 / 1.4
U1 Q	141.956	0.0 / 1.6
U2 G	851.735	1.0 / 1.4
U2 Q	1277.603	0.0 / 1.6

TT1 · Section A · S460 transfer truss · Governing members 1

Axial member utilisation = max(Nc/Pc, Nt/Pt). Effective lengths: in-plane = node spacing; out-of-plane = chord restraint spacing or full web length.

Top / bottom out-of-plane length = 3.00 / 3.00 m. Strong axis is in the truss plane.

Member / section	Actions / resistance (kN)	Slenderness / check
TC4 · Top chord UC 305x305x226 L=3.00 m; py=440	4369.54 / 10868.25 compression 0.00 / 12760.00 tension	λ_x/y 20.83 / 37.13 curves b/c; 40.20% OK
BC3 · Bottom chord UB 457x280x177 L=3.00 m; py=440	0.00 / 8582.19 compression 4369.54 / 10032.00 tension	λ_x/y 14.93 / 44.25 curves a/b; 43.56% OK
D4 · Diagonal UC 305x305x226 L=3.39 m; py=440	0.00 / 10395.84 compression 4941.90 / 12760.00 tension	λ_x/y 23.56 / 41.99 curves b/c; 38.73% OK
V5 · Vertical UC 203x203x127 L=1.58 m; py=440	2319.88 / 6517.73 compression 0.00 / 7172.00 tension	λ_x/y 16.27 / 28.82 curves b/c; 35.59% OK

Geometry: British Steel UB datasheet 19-07-2024 / UC datasheet 10-07-2023, BS EN 10365. S460 grade is a design assumption; supplier availability is not inferred.